

Maximizing Online Retail Revenue

This project aims to quantify the impact of promotions and identify optimal pricing strategies for a specific product category to maximize revenue in online retail. We'll leverage historical sales data to uncover insights into customer behavior and promotional effectiveness.

E-COMMERCE

DATA ANALYSIS

REVENUE OPTIMIZATION



Project Overview: From Data to Dollars

Our objective is to delve deep into sales data to understand the intricacies of promotional impact and pricing. By the end of this analysis, we will be equipped with actionable insights to optimize promotional strategies and pinpoint the most effective price points for our products.

01

Data Loading & Integrity

Initial checks for quality and structure.

02

Feature Engineering

Creating key metrics like Price Paid and Promotion Type.

03

Exploratory Data Analysis

Visualizing trends, discounts, and revenue correlations.

04

Price Elasticity Analysis

Understanding demand responsiveness to price changes.

05

Promotion Lift Analysis

Quantifying the impact and profitability of discounts.

06

Regression Analysis

Modeling factors influencing units sold.

A close-up photograph of a person's hands reviewing a large data table. The table has many columns and rows, with some cells highlighted in yellow and others in red. The person's left hand is resting on the table, and their right hand is holding a pen, ready to write. The table is placed on a light-colored surface, possibly a desk.

1) Load Data and Initial Integrity Check

The first critical step involves loading our sales data and ensuring its reliability. We'll be working with a CSV file located at `/content/cleaned_data.csv`.

Subtask: Load the sales data from the file into a pandas DataFrame. Perform initial data integrity checks such as inspecting for missing values, duplicate entries, and incorrect data types. Display summary statistics and the first few rows of the DataFrame to understand its structure and content.

These initial checks are vital to avoid errors in subsequent analyses. By understanding the data's structure and cleanliness, we can ensure our insights are built on a solid foundation.



2) Feature Engineering: Building Richer Data Data

To gain deeper insights, we need to enrich our dataset with new, strategically derived features. These features will allow us to analyze sales and promotions from various angles.

Subtask: Create new features: `Price_Paid`, `Promotion_Type`, and `Season`.

We will convert 'InvoiceDate' to datetime objects to enable time-based calculations. 'Price_Paid' will be derived from 'UnitPrice' and 'Discount', providing the actual price paid by the customer. 'Promotion_Type' will categorize discounts into meaningful ranges, while 'Season' will capture seasonal buying patterns based on the month.

- ❏ Verifying the first few rows of the updated DataFrame will confirm the successful creation of these new features.

3) Exploratory Data Analysis: Sales Trend & Trend & Seasonality

Understanding sales trends over time is crucial for identifying recurring patterns and seasonality. This helps in predicting future performance and planning promotional campaigns.

Subtask: Plot the sales trend over time to visualize seasonality and overall patterns. The plot should include a legend and appropriate labels for axes. Analyze the visual output to identify any recurring seasonal peaks or troughs.

Visualizing sales data allows us to quickly spot anomalies, growth periods, and cyclical behaviors that might not be obvious from raw numbers alone. We expect to see typical retail seasonal bumps around holidays and specific shopping seasons.



Discount Impact: Units Sold vs. Discount Rate

To understand how customers react to price reductions, we need to analyze the relationship between discount rates and the quantity of units sold.

Subtask: Generate a scatter plot or line plot to visualize the correlation between `Discount_Rate` and `Units_Sold`. This will help understand how units sold respond to different discount levels. Ensure the plot has a legend and clear axis labels.

A clear scatter plot using 'Discount' on the x-axis and 'Quantity' on the y-axis will reveal the elasticity of demand. We are looking for trends where higher discounts lead to significantly increased sales, indicating a positive response to promotions.

Discount Impact: Revenue vs. Discount Rate

While increased units sold is good, the ultimate goal is revenue maximization. This analysis will show how discounts influence overall revenue, not just volume.

Subtask: Generate a scatter plot or line plot to visualize the correlation between `Discount_Rate` and `Revenue`. This will help understand how revenue responds to different discount levels. Ensure the plot has a legend and clear axis labels.

By plotting 'Discount' against 'Revenue', we can identify the discount sweet spot where sales volume increases enough to offset the price reduction, leading to the highest possible revenue. This plot might reveal a peak, indicating an optimal discount level.



5) Promotion Lift Analysis: Quantifying Discount Effectiveness

Beyond simple correlations, we need to understand the true "lift" that promotions provide. This involves comparing promotional sales to a baseline of non-promotional sales.

Subtask: Group the data by `Discount_Rate` brackets. For each bracket, calculate the incremental lift in units sold compared to 'No Discount' days. Also, calculate the incremental profit for each bracket to determine if the extra units sold compensate for the reduced price.

We'll meticulously categorize discounts, establish a baseline from 'No Discount' sales, and then compute the incremental units sold and profit for each promotional tier. This analysis will definitively tell us which promotions are genuinely profitable.



6) Simple Regression Analysis: Modeling Sales Drivers

To predict sales and understand the influence of various factors, we will build a linear regression model. This model will quantify the impact of price, seasonality, and holidays on units sold.

Subtask: Implement a simple linear regression model using `sklearn` with `Units_Sold` as the dependent variable and `Price_Paid`, `Month`, and `IsHoliday` as independent variables. Quantify the impact of price changes on units sold, controlling for seasonality. Provide the model coefficients and their interpretation.

First, we'll create the `IsHoliday` feature by identifying Christmas and New Year's Day. Then, we'll define our target (`Quantity`) and features (`Price_Paid`, `Month`, `IsHoliday`) to train the model. The coefficients will reveal the magnitude and direction of each variable's influence on sales.